A TREE AUTOMATON WITH INFINITELY MANY ROGUE FIXED POINTS UNDER EVERY SUPERCRITICAL OFFSPRING LAW

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Abstract. Johnson, Podder and Skerman asked whether a tree automaton with finitely many, but at least three, states can have infinitely many fixed points of its automaton distributional map. Under binary branching an example with three states is implicit in earlier work of Rozikov and Zada on quadratic stochastic operators. We show that the phenomenon holds for every supercritical offspring law, Poisson laws included. For every $k \geq 2$ there is an automaton on $k+1$ states whose fixed points, for an offspring law with generating function G and mean greater than one, are one point mass together with the $(k-1)$-dimensional simplex of laws that give the distinguished state mass s_*, where $G'(s_*) = 1$. Every point of the simplex is rogue, with no moment assumption on the offspring law; for the points other than the k vertices this has a short self-contained proof.

1. Introduction

Let Σ be a finite set of states. Following [4], a *tree automaton* is a map $A\colon \mathbb{N}_0^{\Sigma} \to \Sigma$: the state of a parent is $A(\vec{n})$, where $\vec{n} = (n_\sigma)_{\sigma\in\Sigma}$ counts its children in each state. Let χ be a law on $\mathbb{N}_0$ with $\chi(\{2, 3, \dots\}) > 0$, the standing assumption of [4, Section 1.2], and let D be the simplex of laws on Σ. The *automaton distributional map* $\Psi\colon D \to D$ sends x to the law of $A(\vec{n})$, where the root has $N \sim \chi$ children whose states are independent with law x [4, Section 1.5]. An *interpretation* of A under $\mathrm{GW}(\chi)$ is a measurable map ι from locally finite, ordered, rooted trees (with their Borel σ-field) to Σ such that, for almost every $\mathrm{GW}(\chi)$ tree t, the colouring $v \mapsto \iota(t(v))$ is compatible with A; the law of $\iota(T)$ is then a fixed point of Ψ, and a fixed point that does not arise this way is called *rogue* [4, Sections 1.4 and 1.5].

Johnson, Podder and Skerman remark that the case $|\Sigma| = 2$ can be ruled out by arguing that the automaton distributional map is analytic. The argument needs a hypothesis on χ: Remark 2.1 gives a two-state automaton whose map is the identity. Fixed points of such maps are solutions of recursive distributional equations in the sense of the survey of Aldous and Bandyopadhyay [1], and two-state examples arise as the outcome probabilities of the games on Galton–Watson trees studied by Holroyd and Martin [3]. Johnson, Podder and Skerman asked [4, Section 6]: when $3 \leq |\Sigma| < \infty$, can an automaton have infinitely many fixed points?

For binary branching the answer was in effect already known. Rozikov and Zada [5, Theorem 2(iv) and equation (21)] found a segment of fixed points for a family of quadratic stochastic operators on the two-dimensional simplex; Ganikhodzhaev, Mukhamedov and Rozikov [2] survey these operators. At their parameters $a = b = 0$, $c = 1$ the operator is $(x, y, z) \mapsto (2xz,\ 2yz,\ z^2+(x+y)^2)$; its heredity coefficients are 0 or 1, so it is the automaton distributional map of a deterministic three-state automaton under $\mathrm{GW}(\delta_2)$, namely the automaton A_2 below,

2020 *Mathematics Subject Classification.* Primary 60J80; Secondary 68Q45, 60E05.

Key words and phrases. Tree automaton, Galton–Watson tree, recursive distributional equation, fixed point, interpretation, rogue fixed point.

and its fixed segment is $x + y = 1/2$. What follows extends this example to every supercritical offspring law ([4] treats general offspring laws, and most of its examples are Poisson) and proves that every point of the simplex is rogue.

Fix $k \geq 2$ and let $\Sigma_k = \{a, b_1, \dots, b_k\}$. Call a the *even* state and the b_i the *odd* states. Let A_k be the automaton whose output is b_i when the vertex has exactly one odd child and that child is in state b_i, and a otherwise:

$$A_k(\vec{n}) = \begin{cases} b_i & \text{if } n_{b_i} = 1 \text{ and } n_{b_j} = 0 \text{ for } j \neq i, \\ a & \text{otherwise.} \end{cases}$$

The number n_a of even children plays no role, and a leaf is in state a. On a vertex with two children, A_k is the symmetric rule

$$g(a, a) = a, \qquad g(b_i, b_j) = a, \qquad g(a, b_i) = g(b_i, a) = b_i,$$

and on a vertex with one child it copies that child.

Write $G(s) = \sum_{n \geq 0} \chi(n) s^n$ for the generating function of χ and $m = G'(1) \in (0, \infty]$ for its mean. Under the standing assumption $\chi(\{2, 3, \dots\}) > 0$, G' is continuous and strictly increasing on $[0, 1)$, with $G'(0) = \chi(1) < 1$. So when $m > 1$ there is a unique $s_* \in (0, 1)$ with $G'(s_*) = 1$.

Theorem 1.1. *If $m > 1$, the fixed points of the automaton distributional map of A_k under $\mathrm{GW}(\chi)$ are exactly the point mass δ_a and the laws x with $x(a) = s_*$. The latter form a $(k-1)$-dimensional simplex, whose relative interior consists of laws of full support. If $m \leq 1$, the only fixed point is δ_a.*

Corollary 1.2. *For $\chi = r\delta_1 + (1 - r)\delta_2$ with $0 \leq r < 1$, $s_* = 1/2$; for $k = 2$ the three-state automaton A_2 then has the line of fixed points $(\frac{1}{2}, t, \frac{1}{2} - t)$, $0 \leq t \leq \frac{1}{2}$, in the order (a, b_1, b_2). For the Poisson law with mean $\lambda > 1$, $s_* = 1 - (\log \lambda)/\lambda$.*

Proof. In the first case $G'(s) = r + 2(1 - r)s$, and in the second $G'(s) = \lambda e^{\lambda(s-1)}$; solve $G'(s) = 1$. □

Theorem 1.3. *Let $m > 1$ and let ι be an interpretation of A_k under $\mathrm{GW}(\chi)$. Then the law of $\iota(T)$ is δ_a or $s_*\delta_a + (1 - s_*)\delta_{b_i}$ for some i. Hence every fixed point other than these $k + 1$ laws is rogue; in particular so is every fixed point of full support.*

Theorem 1.4. *If $m > 1$, then δ_a is the only fixed point of A_k under $\mathrm{GW}(\chi)$ that admits an interpretation.*

Proposition 1.5. *Under $\mathrm{GW}(\delta_2)$ the only interpretation of A_k, up to almost sure equality, is the constant map with value a. Hence every fixed point with $x(a) = 1/2$ is rogue.*

Thus for every supercritical offspring law the fixed-point set contains a continuum of rogue laws. Theorem 1.4 contains Theorem 1.3 and Proposition 1.5; we keep them because their proofs are elementary, while the proof of Theorem 1.4 goes through the spine argument of [4, Section 4].

2. Proofs

Proof of Theorem 1.1. Let x be a law on Σ_k and write $s = x(a)$. Given $N = n$, the children are independent with law x, and the root is in state b_i exactly when one child is b_i and the other $n - 1$ are a; this has probability $n\, x(b_i) s^{n-1}$. Summing over n,

$$\Psi(x)(b_i) = x(b_i)\, G'(s), \qquad \Psi(x)(a) = 1 - (1 - s)\, G'(s),$$

for $s < 1$, where $G'(s)$ is finite. (A root with no children is in state a, and the term $n = 0$ contributes nothing to the first formula.)

If $x(b_i) = 0$ for every i, then $x = \delta_a$ and $\Psi(x) = \delta_a$. Otherwise $s < 1$, and a fixed point must satisfy $G'(s) = 1$, from the coordinate of any b_i with $x(b_i) > 0$. Conversely, if $G'(s) = 1$ then $\Psi(x)(b_i) = x(b_i)$ for every i and $\Psi(x)(a) = 1 - (1 - s) = s$, so x is fixed. For $s < 1$ we have $G'(s) < G'(1) = m$, so $G'(s) = 1$ has no solution when $m \le 1$ and exactly the solution s_* when $m > 1$. The laws with $x(a) = s_*$ are $s_*\delta_a + (1 - s_*)\mu$ with μ any law on $\{b_1, \dots, b_k\}$, a $(k-1)$-simplex, and x has full support exactly when μ does. □

For $\chi = \delta_2$ and $k = 2$, writing (x_0, x_1, x_2) for the masses of (a, b_1, b_2), the map is

$$\Psi(x_0, x_1, x_2) = (x_0^2 + (x_1 + x_2)^2,\ 2x_0x_1,\ 2x_0x_2),$$

and the line in Corollary 1.2 can be checked by substitution.

Proof of Proposition 1.5. Under $\mathrm{GW}(\delta_2)$ the tree is almost surely the infinite binary tree t_*, and every subtree $t_*(v)$ equals t_*. An interpretation ι therefore colours every vertex of t_* with the single value $c = \iota(t_*)$, and compatibility at a vertex with two children in state c requires $c = g(c, c)$. This holds for $c = a$ and fails for $c = b_i$, since $g(b_i, b_i) = a$. Conversely the constant map a is compatible at every vertex. So the law of $\iota(T)$ is δ_a, and every fixed point with $x(a) = 1/2$ is rogue. □

Proof of Theorem 1.3. Let x be the law of $\iota(T)$; it is a fixed point [4, Section 1.5], so by Theorem 1.1 either $x = \delta_a$ or $s := x(a) = s_*$. Assume the latter. Call a tree t *odd* if $\iota(t) \neq a$ and *even* otherwise, and let O be the set of odd trees, so $\mathbb{P}(T \in O) = 1 - s_* > 0$. Almost surely the colouring $v \mapsto \iota(T(v))$ is compatible with A_k at every vertex. By the definition of A_k, an odd vertex v then has exactly one odd child $c(v)$, and $\iota(T(c(v))) = \iota(T(v))$.

For an odd tree t with this property at its root, let $\theta t = t(c(o))$ be the subtree of the odd child and let $\xi(t)$ record the rest of the first generation: the number n of children of the root, the position j of the odd child among them, and the $n - 1$ even subtrees of its siblings, in order. Given $N = n$ the subtrees of the children of the root are independent $\mathrm{GW}(\chi)$ trees, and whether each is odd is decided by that subtree alone. Hence, for Borel sets B of trees and C of decorations,

$$\mathbb{P}(T \in O,\ \theta T \in B,\ \xi(T) \in C) = \mathbb{P}(T \in B \cap O)\, \nu(C),$$

where $\nu(C) = \sum_n \chi(n) \sum_{j=1}^{n} \mathbb{P}((n, j, T_1, \dots, T_{n-1}) \in C,\ T_1, \dots, T_{n-1} \notin O)$ with $T_1, T_2, \dots$ independent $\mathrm{GW}(\chi)$ trees. The total mass of ν is $\sum_n \chi(n)\, n\, s_*^{n-1} = G'(s_*) = 1$. Dividing by $\mathbb{P}(T \in O)$: under $\mathbb{P}(\cdot \mid T \in O)$, θT is odd with the conditional law of T, and $\xi(T)$ has law ν and is independent of θT.

Set $T_0 = T$, $T_{r+1} = \theta T_r$ and $\xi_r = \xi(T_r)$; these are defined almost surely on $\{T \in O\}$, since every vertex of the odd ray $o, c(o), c(c(o)), \dots$ is odd. Iterating the previous paragraph, under $\mathbb{P}(\cdot \mid T \in O)$ the ξ_r are independent with law ν, and $\xi_0, \dots, \xi_{r-1}$ are independent of T_r. The tree T_r is the odd ray from its root with the decorations $\xi_r, \xi_{r+1}, \dots$ hung along it; its first h generations are determined by $\xi_r, \dots, \xi_{r+h-1}$, so T_r is a measurable function of $(\xi_r, \xi_{r+1}, \dots)$. Fix i. Since ι is constant along the odd ray, $\mathbf{1}\{\iota(T_0) = b_i\} = \mathbf{1}\{\iota(T_r) = b_i\}$ almost surely, and the right side is $\sigma(\xi_r, \xi_{r+1}, \dots)$-measurable. So $\{\iota(T) = b_i\}$ agrees almost surely with an event of the tail σ-field of the independent sequence (ξ_r), and by Kolmogorov's zero-one law $\mathbb{P}(\iota(T) = b_i \mid T \in O) \in \{0, 1\}$. Thus x restricted to $\{b_1, \dots, b_k\}$ is concentrated on a single state. □

Proof of Theorem 1.4. Merge $b_1, \dots, b_k$ into one state b. The result is the two-state automaton B that outputs b when exactly one child is in state b, and a otherwise. If ι interprets a fixed

point x of A_k with $x \neq \delta_a$, then composing ι with the merging map interprets the fixed point $\bar{x} = (s_*, 1-s_*)$ of B. So it suffices to show that $\bar{x}$ is rogue. By [4, Theorem 1.7] this holds if the pivot tree of $\bar{x}$ is supercritical, provided χ has a finite logarithmic moment; we first check that for B the moment is not needed, and then compute the pivot tree.

The direction of [4, Theorem 1.7] used here is proved in [4, Section 4]. The logarithmic moment is used substantively only in [4, Lemma 4.14], through [4, Lemma 4.13], and [4, Lemma 4.15] inherits it from there. In [4, Lemma 4.14] it serves to bound the number of children of the root of T^{0s} and of T^{1s}, where T^{ℓ} is the random state tree conditioned on the label ℓ of its root, and $\ell = 0s$ or $1s$ means that the root has the colour 0 or 1 and starts an infinite path of pivotal vertices [4, Section 4]. For B, with the colours a and b in the roles of 0 and 1, write $s = s_*$ and $p = 1 - s_*$, and let N be the number of children of the root. Then $\mathbb{P}(N = n,\ \text{root in state } b) = \chi(n)\, n\, p\, s^{n-1}$, since the root then has exactly one child in state b. A root labelled $0s$ is in state a and has a pivotal child, so, as shown below, it has either zero or exactly two children in state b, and $\mathbb{P}(N = n,\ \text{label } 0s) \le \chi(n)\big(s^n + \binom{n}{2} p^2 s^{n-2}\big)$. The labels $0s$ and $1s$ have positive probability when the pivot tree is supercritical [4, Proposition 2.6(c)], so dividing by $\mathbb{P}(0s)$ and $\mathbb{P}(1s)$,

$$\mathbb{P}(N = n \mid 0s) \le \frac{\chi(n)\big(s^n + \binom{n}{2} p^2 s^{n-2}\big)}{\mathbb{P}(0s)}, \qquad \mathbb{P}(N = n \mid 1s) \le \frac{\chi(n)\, n\, p\, s^{n-1}}{\mathbb{P}(1s)},$$

and both are at most $C\, n^2 s^{n-2}$ for a constant C. As $s < 1$, the root degrees D_{0s} and D_{1s} of T^{0s} and T^{1s} have all moments; put $L = \max\{\mathbb{E}D_{0s}, \mathbb{E}D_{1s}\} < \infty$.

With these bounds the conclusion of [4, Lemma 4.14] holds without the moment assumption. We give the argument in full, because the end of the proof in [4] conditions a spine subtree on the whole spine σ-field, which [4, Proposition 4.9(c)] does not cover. In the notation of [4, Section 4], let $\mathcal{H}_i$ be the information revealed by exploring the spine $V_0, \dots, V_{i-1}$, revealing at each explored spine vertex all its children with their labels but nothing below the children off the spine. The weights $W_0, \dots, W_{i-1}$ are $\mathcal{H}_i$-measurable. Given $\mathcal{H}_i$, the unswitched subtree at V_i has law RST_{0s} or RST_{1s} according to its label, so $\mathbb{E}[\deg V_i \mid \mathcal{H}_i] \le L$, and the proof of [4, Lemma 4.12] goes through with $\mathcal{H}_i$ in place of the spine σ-fields, giving $\mathbb{E}[W_0 \cdots W_{i-1}] \le c^{\lfloor i/2 \rfloor}$ for some $c < 1$. Since $N(V_i, T^{0s}, \omega_*^{0s \to 1s}) \le \deg V_i$,

$$\mathbb{E} \sum_{i \ge 0} W_0 \cdots W_{i-1}\, N(V_i, T^{0s}, \omega_*^{0s \to 1s}) \le \sum_{i \ge 0} \mathbb{E}\big[W_0 \cdots W_{i-1}\, \mathbb{E}[\deg V_i \mid \mathcal{H}_i]\big] \le L \sum_{i \ge 0} c^{\lfloor i/2 \rfloor} < \infty,$$

so the series of [4, Lemma 4.14] is finite almost surely. The rest of the proof of [4, Theorem 1.7], that a supercritical pivot tree makes the fixed point rogue, then goes through without the moment assumption on χ.

In the random state tree of $\bar{x}$ [4, Section 1.6], a vertex is pivotal when changing its colour, and recolouring its ancestors by B, changes the colour of the root; the pivot tree is a two-type Galton–Watson tree, the type being the colour [4, Proposition 2.4]. Given the colour of a vertex, its children are N independent colours with law $\bar{x}$, conditioned on B producing that colour. A vertex in state b has exactly one child in state b, and changing any one child's colour leaves it with zero or two such children, so every child is pivotal. A vertex in state a with no child in state b has all its children pivotal, all in state a; one with exactly two children in state b has those two pivotal and no others; one with three or more has none. Taking expectations, with $\mathbb{P}(b) = pG'(s) = p$ and $\mathbb{P}(a) = s$, the mean offspring matrix of the pivot tree, rows and columns ordered (b, a), is

$$M = \begin{pmatrix} 1 & s\, G''(s) \\ p^2 G''(s)/s & 1 \end{pmatrix} :$$

from b, one pivotal child of type b and $\sum_n \chi(n)\, n(n-1)s^{n-1} = sG''(s)$ of type a on average; from a, $\sum_n \chi(n)\, n\, s^n/s = G'(s) = 1$ of type a and $2\sum_n \chi(n)\binom{n}{2}p^2 s^{n-2}/s = p^2 G''(s)/s$ of type b. Since $\chi(\{2,3,\dots\}) > 0$ we have $G''(s) \in (0,\infty)$, and the spectral radius of M is $1 + pG''(s) > 1$. □

Remark 2.1. Two states can suffice when χ has infinite mean. Let $\chi(n) = 1/(n(n-1))$ for $n \geq 2$, which has infinite mean and a finite logarithmic moment, and let the two-state automaton on $\{0,1\}$ output 1 exactly when at most one child is in state 0. If each child is in state 1 with probability p, then

$$\Psi(p) = \sum_{n\geq 2} \frac{p^n + n(1-p)p^{n-1}}{n(n-1)} = \sum_{n\geq 2} \Big(\frac{p^{n-1}}{n-1} - \frac{p^n}{n} \Big) = p,$$

so every law on $\{0,1\}$ is fixed. If instead $\chi(0) > 0$ and G is entire, as for Poisson laws, three states are needed: Ψ is then analytic on a neighbourhood of $[0,1]$, and it is not the identity, since a leaf in state 0 gives $\Psi(1) \leq 1 - \chi(0)$ and a leaf in state 1 gives $\Psi(0) \geq \chi(0)$; so Ψ has finitely many fixed points in $[0,1]$.

The argument in the proof of Theorem 1.4 applies to any two-state automaton and nondegenerate fixed point: under the other hypotheses of [4, Theorem 1.7], including full support of the fixed point, a supercritical pivot tree makes the fixed point rogue whenever the root of T^{0s} and of T^{1s} has a number of children with a finite logarithmic moment, whatever χ is. (With only a logarithmic moment, the exponential decay of $W_0 \cdots W_{i-1}$ in [4, Lemma 4.14] and the Borel–Cantelli lemma replace the bound by L.) For Poisson offspring with mean $\lambda > 1$ the spectral radius in the proof of Theorem 1.4 is $1 + \log\lambda$.

Acknowledgements and funding

The final manuscript was audited by Claude Opus 5.5 (Anthropic), which checked the proofs and the cited results against the sources. The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [7] and working together inside the Brain framework [6], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [8], the Math Brain repository [7] and the Brain framework repository [6].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Code and data availability

The code and data for this paper are at https://github.com/yspennstate/automaton-fixed-simplex. The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [7].

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, a project on the use of AI systems for research mathematics. An OpenAI GPT model proposed the first example, a four-state automaton with a line of fixed points, and in later sessions pointed to the example of Rozikov and Zada, the reduction used for Theorem 1.4, the two-state example in Remark 2.1 and the form of the series estimate in the proof. Claude Opus 5.5 (Anthropic) found the automaton presented here, extended it to general offspring laws, removed the logarithmic moment assumption from Theorem 1.4 and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

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